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Mortality Associated With Tracheostomy Complications in the United States: 2007–2016

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Objectives/Hypothesis: To investigate patterns of tracheostomy-associated death in the United States.

Study Design: Retrospective database review.

Methods: We used Multiple Cause-of-Death data from the Centers for Disease Control and Prevention to determine cumulative national mortality associated with tracheostomy complications in the United States from 2007 to 2016. Using *International Classification of Diseases, Tenth Revision* data, we investigated how frequently tracheostomy-related complications were reported in cause of death data. We then compared the characteristics of patients with tracheostomy-related mortality to patients reported to have died of other causes.

Results: Over the 10-year period studied, we identified 623 tracheostomy-related deaths (537 adults and 86 children) out of 25,587,306 total deaths reported. Although absolute mortality was higher in adults, the reported base rate of tracheostomy complication–associated mortality was tenfold higher in children. Most tracheostomy-related deaths occurred in a hospital facility (74.5% in adults). Deaths associated with tracheostomy complications were significantly more common for African American children (odds ratio [OR]: 2.02, 95% confidence interval [CI]: 1.26-3.24) and adults (OR: 1.59, 95% CI: 1.29-1.96) or in Hispanic adults (OR: 1.42, 95% CI: 1.06-1.89). Deaths related to a tracheostomy complication more commonly occurred on the weekend (OR: 1.24, 95% CI: 1.04-1.49) and in the most recent 2-year period (OR: 1.31, 95% CI: 1.03-1.68). Adults with a bachelor's, master's or doctorate degree were significantly less likely to have mortality associated with a tracheostomy-related complication (OR: 0.54, 95% CI: 0.39-0.75).

Conclusions: Tracheostomy-related complications were implicated in the deaths of a significant number of individuals. Several demographic groups had increased likelihood of tracheostomy-related mortality. Future research is necessary to develop targeted interventions to decrease harm.

Key Words: Tracheostomy, tracheotomy, mortality, patient safety, quality improvement.

Level of Evidence: NA

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INTRODUCTION

Tracheostomy is often performed in patients who are medically fragile and ill-equipped to respond to emergencies due to aphonic state, neurological impairment, or limited functional status. Tracheostomy carries a risk for rare, catastrophic complications such as mucus plugging, accidental tube dislodgement, or tracheoinnominate artery fistula, any of which can lead to death if not prevented or rapidly treated.[1–3] Most catastrophic events involve either a lack of preventive measures or failure to rescue expeditiously.[2] These tracheostomy-related events are largely preventable, with robust systems to ensure unified and consistent care of tracheostomy patients.[4,5]

Although survey-based studies have provided critical insights into causal factors involved in catastrophic tracheostomy complications,[6] few large-scale studies have been conducted. The National Confidential Enquiry Into Patient Outcomes and Death publication, *On the Right Trach*, provided insights into etiologies of tracheostomy-related death in the United Kingdom.[7] No such data have been reported in the United States. Deeper understanding is needed of those factors predisposing to tracheostomy-related death to inform future quality improvement efforts.

Our objectives were: 1) to determine the reported incidence of tracheostomy-related death in the US over a 10 year period and 2) to identify to whom these catastrophic complications happen, where these deaths occur, and when these deaths happen to inform future quality improvement efforts.

MATERIALS AND METHODS

Data Source

We used Multiple Cause-of-Death data from the Centers for Disease Control and Prevention (CDC) to identify instances of

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J.D.C. was responsible for conception and design; acquisition, analysis, and interpretation of data; drafting and revising critically for important intellectual content. E.M.G. and M.J.B. were responsible for conception and design, analysis and interpretation of the data, and revising critically for important intellectual content.

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tracheostomy-related death in the United States from 2007 to 2016. This database collects mortality data on all deaths that occur within the United States. Each record is abstracted from death certificates that require the medical certifier to enter the chain of events, diseases, injuries or complications that directly caused the death.[8] A standard template and example death certificate can be found on the CDC website.[8] Death certificates are filed in vital statistics offices in each state where this information is abstracted into *International Classification of Diseases, Tenth Revision* (ICD-10) codes, and demographic data are recorded. The database coding system used by the CDC does not allow tracheostomy complication and other procedure-specific codes to be listed on the final line of the death certificate reserved for underlying cause. Therefore, a biological condition, such as respiratory failure or laryngeal cancer, that led to the tracheostomy must be listed. The Multiple Cause of Death data are deidentified and publicly available. Analysis of this deidentified data of deceased individuals is not regulated as human subjects research and does not require oversight or institutional review board approval.

Subjects and Variables

We identified deaths associated with tracheostomy complication if ICD-10 code J95.0 was recorded as a cause of death on the death certificate (see Supporting Table 1 in the online version of this article). This code corresponds to tracheostomy complications and encompasses all common complications with tracheostomy as well as unspecified tracheostomy complications. The terms *death associated with tracheostomy* and t*racheostomy-related death* are used interchangeably to refer to individuals where tracheostomy complication was reported as a cause of death. We compared characteristics of individuals with tracheostomy-related death to the rest of the population who died of other causes during this period. We use the term *deceased general population* (or simply *general population*) to describe our comparison group, consisting of all deceased individuals reported in the same period without a reported tracheostomy complication as a sequential condition resulting in death. We subdivided the study cohort and comparison cohort into children (under 18 years old) and adults (18+ years old) because indications for tracheostomy, types of tracheostomy tubes, and overall features are different in these populations.[9] The location of death was divided as an inpatient, in an emergency department or outpatient setting, dead on arrival to a hospital, nursing home/long-term care facility, hospice, at the decedent's home, other, or unknown. Marital status at the time of death is identified by the CDC as married, married but separate, widowed, divorced, never married, or unknown. The term widowed is intended to apply to individuals of any gender who have lost a spouse to death.

Statistical Analysis

For descriptive analysis, trends in the mortality rate were plotted for each day of the week, month, and year. Multivariate logistic regression models were developed in a forward, stepwise fashion to compare characteristics in the group that died of a tracheostomy complications versus the general population. We separated children and adults, and selected patient-level variables based on clinical relevance. In children, our final model included eight variables with a C statistic of 0.72. In adults, our final model included 10 variables with a C statistic of 0.81. Hosmer-Lemeshow goodness-of-fit statistics showed the model was well calibrated. We report the odds ratio (OR) with 95% confidence interval (CI) for all comparisons. We performed statistical analyses with SPSS version 24 (IBM, Armonk, NY).

RESULTS

We identified 623 instances of death associated with a tracheostomy complication among the 25,587,306 deaths in the United States from 2007 to 2016. Of these deaths, 86 (14%) of the tracheostomy-related deaths occurred among the 396,288 deaths in children, and 537 (86%) of the tracheostomy-related deaths occurred among the 25,169,470 deaths in adults. Tracheostomy-related deaths accounted for 0.022% of all deaths in children and 0.0022% of all deaths in adults. These incidences corresponded to an approximately tenfold higher base rate of reported pediatric versus adult tracheostomy-related death.

Pediatric Tracheostomy-Related Mortality

In children, the majority of deaths (59.3% of cases) occurred in males (Table I.). Significant racial disparities existed for children, with 40.7% of the tracheostomy-related deaths occurring in African Americans, whereas only 27.8% of individuals within the deceased general population were African American. Most of the pediatric tracheostomy-related deaths occurred in the hospital (76.1%). The timing of tracheostomy-related deaths in relation to the day of the week, month, and year was nonrandom (Fig. 1).

Table II shows the underlying cause of death reported for children in the general population as compared to tracheostomy-related deaths. Chronic lower respiratory tract disease was reported as the primary cause of death in 20% of all tracheostomy-related deaths regardless of age, versus only 2% of pediatric and 8% of adult deaths in the deceased general population. Furthermore, complications of medical and surgical care were noted as the primary cause of death in 24% of tracheostomy-related deaths versus only 0.1% of all other deaths.

Adult Tracheostomy-Related Mortality

In adults, the average age at death from a tracheostomy complication was 60.7 years versus 74.0 years in the deceased general population (Table I). Significant racial and ethnic disparities existed for tracheostomy mortality reported for adults. African Americans accounted for 24.4% of tracheostomy-related deaths versus 11.5% in the adult general population, and Hispanics accounted for 10.8% of deaths versus 6% in the adult general population. The hospital was the most common location for tracheostomy-related deaths (74.5%). Tracheostomy-related deaths were more common on Saturday or Sunday (33.1%) (Fig. 2). The yearly number of deaths from a tracheostomy complication varied from a nadir of 35 deaths in 2012 (30 in adults) to a high of 113 deaths in 2016 (98 in adults). In adults, the underlying cause of death was more frequently compared with the general population for malignant neoplasm of the oral cavity/pharynx (3.4% vs. 0.4%), chronic lower respiratory tract disease (19.6% vs. 7.8%), or a complication of medical and surgical care (24.6% vs. 0.1%).

TABLE I.
Demographic Characteristics of Pediatric and Adult Cohorts.

Variable	Pediatrics		Adults	
	Tracheostomy Complication, n = 86	Deceased General Population, n = 396,202	Tracheostomy Complication, n = 537	Deceased General Population, n = 25,168,933
Age				
Mean (SD), yr,	4.7 (4.9)	4.2 (5.5)	60.7 (15.7)	74 (16.5)
<2 years	34 (39.5)	266,807 (67.3)		
2–5 years	23 (26.7)	25,762 (6.5)		
5–17 years	29 (33.7)	103,633 (26.2)		
18–44 years			76 (14.2)	1,455,956 (5.8)
45–64 years			229 (46.6)	5,082,936 (20.2)
65–80 years			178 (33.1)	7,206,116 (28.6)
80+ years			54 (10.1)	11,441,925 (45.4)
Gender, no. (%)				
Female	35 (40.7)	167,531 (42.3)	243 (45.3)	12,570,232 (49.9)
Male	51 (59.3)	228,671 (57.7)	294 (54.7)	12,616,701 (50.1)
Race, no. (%)				
White	47 (54.7)	265,065 (66.9)	386 (71.9)	21,579,683 (85.7)
African American	35 (40.7)	110,150 (27.8)	131 (24.4)	2,896,065 (11.5)
American Indian	1 (1.2)	6,608 (1.7)	7 (1.3)	159,466 (0.6)
Asian or Pacific Islander	3 (3.5)	14,379 (3.6)	13 (2.4)	551,719 (2.2)
Ethnicity, no. (%)				
Non-Hispanic	70 (81.4)	312,895 (79.0)	476 (88.6)	23,619,337 (93.8)
Hispanic	16 (18.6)	80,411 (23.0)	58 (10.8)	1,499,319 (6.0)
Unknown	0 (0)	2,896 (0.7)	3 (0.6)	68,277 (0.3)
Education, no. (%)				
8th grade or less	73 (84.9)	312,533 (78.9)	72 (13.4)	2,988,467 (11.9)
9th–12th grade, no diploma	9 (10.5)	42,976 (10.8)	71 (13.2)	2,869,543 (11.4)
High school graduate or GED	0 (0)	3,047 (0.8)	228 (42.5)	10,561,894 (41.9)
Some college	0 (0)	166 (0)	58 (10.8)	2,591,931 (10.3)
Associate's degree			39 (7.3)	1,660,640 (6.6)
Bachelor's degree			30 (5.6)	2,373,368 (9.4)
Master's degree			14 (2.6)	1,035,160 (4.1)
Doctorate or professional degree			3 (0.6)	323,692 (1.3)
Unknown	4 (4.7)	37,480 (9.5)	22 (4.1)	782,238 (3.1)
Marital status, no. (%)				
Single, never married			130 (24.2)	2,799,011 (11.1)
Married			236 (43.9)	9,527,295 (37.8)
Divorced			94 (17.5)	3,672,379 (14.6)
Widowed			68 (12.7)	8,913,406 (35.4)
Unknown			9 (1.7)	274,842 (1.1)
Location of death, no. (%)				
Hospital, inpatient	50 (58.1)	240,755 (60.8)	305 (56.8)	7,948,818 (31.6)
Hospital, emergency department or outpatient	16 (18.6)	67,231 (17.0)	95 (17.7)	1,656,961 (6.6)
Dead on arrival to hospital	4 (4.7)	7,101 (1.8)	3 (0.6)	141,797 (0.6)
Decedent's home	13 (15.1)	48,606 (12.3)	61 (11.4)	7,077,599 (28.1)
Hospice	0 (0)	1,404 (0.4)	3 (0.6)	1,414,873 (5.6)
Nursing home/long-term care	1 (1.2)	815 (0.2)	59 (11.0)	5,234,426 (20.8)
Other	1 (1.2)	28075 (7.1)	9 (1.7)	1,580,026 (6.3)
Unknown	1 (1.2)	2215 (0.6)	2 (0.4)	132,433 (0.2)
Autopsy, no. (%)				

(Continues)

TABLE I.
Continued

Variable	Pediatrics		Adults	
	Tracheostomy Complication, n = 86	Deceased General Population, n = 396,202	Tracheostomy Complication, n = 537	Deceased General Population, n = 25,168,933
No	50 (58.1)	216,859 (54.7)	408 (76.0)	21,276,223 (84.5)
Yes	29 (33.7)	152,706 (38.5)	76 (14.2)	1,861,110 (7.4)
Unknown	7 (8.1)	26,637 (6.7)	53 (9.9)	2,049,600 (8.1)

GED = general education development test; SD = standard deviation.

Multivariate Analysis of Characteristics Associated With Tracheostomy-Associated Mortality

We used multivariate logistic regression analysis to examine characteristics associated with death caused by a tracheostomy complication in children (Table III). In the pediatric group, age 2 to 5 years (OR: 7.76, 95% CI: 4.53-13.27), age 5 to 17 years (OR: 2.63, 95% CI: 1.57-4.42), and African American race (OR: 2.02, 95% CI: 1.26-3.24) were associated with significantly increased likelihood of tracheostomy-related death. There was no association with Hispanic ethnicity, location of death, day of the week, month of the year, or year.

In adults, age 18 to 44 years and age 45 to 64 years were both significantly associated with tracheostomy-related death, whereas age greater than 80 years was associated with a reduced risk of tracheostomy-related death. African American race and Hispanic ethnicity were both associated with an increased likelihood of tracheostomy-related death. Education of a bachelor's degree, master's degree, or doctorate was associated with decreased likelihood of tracheostomy-related death. Widowed individuals were also less likely to die with a tracheostomy complication compared to the deceased general population. Tracheostomy-related death was more likely to occur in the hospital or in a nursing home/long-term care facility. Tracheostomy-related deaths were more likely to occur on the weekend. There were significantly more tracheostomy-related deaths in the most recent years available, 2015 to 2016, compared with other time periods.

DISCUSSION

Despite the major advances in tracheostomy devices, surgical technique, ventilation, intensive care, and capabilities for treating complex patients in the past century, deaths from tracheostomy complications remain a persistent problem.[2] In the 623 tracheostomy-related deaths analyzed in this study, we identified significant variation in demographic characteristics, timing, and location of deaths that highlight potential opportunities for quality improvement. The findings are particularly troubling, because many of these deaths are in many ways believed to be preventable[2] and appear to preferentially befall certain vulnerable groups.

Tracheostomy is frequently performed in critically ill patients, and the preponderance of deaths in prior studies have been attributed to underlying critical illness. Multi-institutional studies have identified a 19% to 22% in-hospital mortality rate in adults[2,10] and 9% to 39% 2-year mortality rate in children[11] after tracheostomy. Although

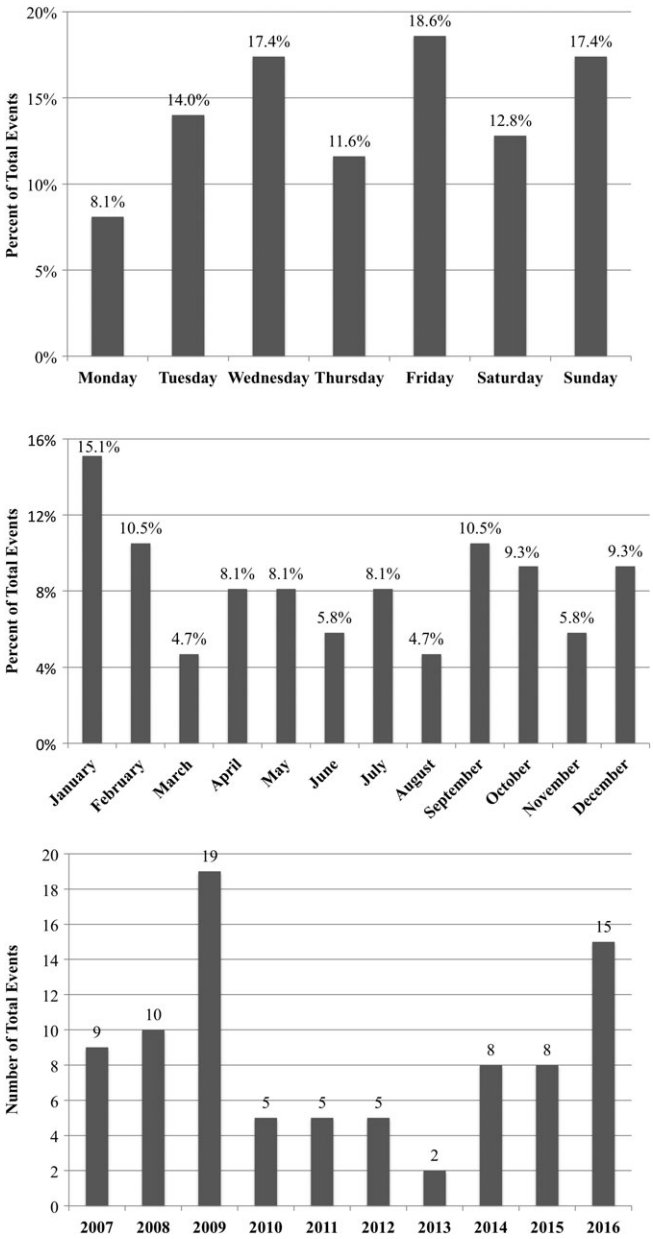

Fig. 1. Day of week, month, and year of deaths related to tracheostomy complications in the pediatric cohort.

TABLE II.
Underlying Cause of Death in Pediatric and Adult Cohorts.*

Variable	Pediatrics		Adults	
	Tracheostomy Complication, n = 86, No. (%)	Deceased General Population, n = 396,202, No. (%)	Tracheostomy Complication, n = 537, No. (%)	Deceased General Population, n = 25,168,933, No. (%)
Malignant neoplasm, oral cavity and pharynx	0 (0)	53 (0)	18 (3.4)	88,273 (0.4)
Malignant neoplasm, larynx and lung	0 (0)	113 (0)	11 (2.0)	1,564,171 (6.2)
Other malignant neoplasms	2 (2.3)	18,703 (4.7)	58 (10.8)	4,287,882 (17.0)
Chronic lower respiratory tract disease	17 (19.8)	7,679 (1.9)	105 (19.6)	1,953,186 (7.8)
Pneumonia/acute lower respiratory tract infection	1 (1.2)	5,147 (1.3)	12 (2.2)	536,336 (2.1)
Other infectious disease	1 (1.2)	9,406 (2.4)	5 (0.9)	684,940 (2.7)
Cardiovascular disease	1 (1.2)	9,450 (2.4)	29 (5.4)	6,717,739 (26.7)
Diabetes mellitus	0 (0)	537 (0.1)	8 (1.5)	739,423 (2.9)
Gastrointestinal or liver disease	0 (0)	620 (0.2)	3 (0.6)	433,656 (1.7)
Renal disease	0 (0)	1,420 (0.4)	4 (0.7)	491,452 (2.0)
Neurologic disease	3 (3.5)	3,824 (1.0)	31 (5.8)	2,467,282 (9.8)
Pregnancy, childbirth and peripartum	0 (0)	133 (0)	0 (0)	11,118 (0)
Congenital malformation	23 (26.7)	185,489 (46.8)	2 (0.4)	39,037 (0.2)
All other diseases	14 (16.3)	60,604 (15.3)	72 (13.4)	3,291,071 (13.1)
Transport accidents	1 (1.2)	29,399 (7.4)	12 (2.2)	376,764 (1.5)
Complications of medical and surgical care	18 (20.9)	586 (0.1)	132 (24.6)	25,994 (0.1)
Other accidents	5 (5.8)	60,263 (15.2)	34 (6.3)	1,426,524 (5.7)
Undetermined	0 (0)	2,776 (0.7)	1 (0.2)	52,085 (0.2)

*This is the disease or injury that initiated the events resulting in death.

no comprehensive prospective data on tracheostomy-related death are available, a national survey of 478 otolaryngologists reported approximately one catastrophic tracheostomy event causing death every 20 years in practice.[6] The authors estimated an annual incidence of approximately 500 tracheostomy-related deaths in the United States.[6] By analyzing data from death certificates, we identified 623 tracheostomy-related deaths over a 10-year period, roughly tenfold lower than this estimate. The varying estimates of the frequency of tracheostomy-related mortality reflect inherent difficulties of such research. The incidence of tracheostomy-related death may be lower than previously thought. Alternatively, underreporting of tracheostomy complications may occur due to a hesitancy to report iatrogenic complications, desire to limit potential liability, complacency with what is perceived to be an unmodifiable level of risk, or a pervasive lack of knowledge about tracheostomy tubes,[12] causing tracheostomy-related complications to go unidentified.

We observed that several groups have an increased likelihood of tracheostomy-related death compared to the general population. Men, African Americans, and Hispanic patients were all at increased risk of death from a tracheostomy complication compared with the deceased general population. Some of these disparities were amplified in children, where 40.7% of deaths occurred in African Americans. Data from the Nationwide Inpatient Sample suggest that mortality after tracheostomy from any cause is also higher in African Americans.[10] The reasons for these disparities are difficult to discern. Future work may explore the impact of fragmented care, as work in head and neck surgery patients has highlighted the prevalence and adverse outcomes associated with care fragmentation[13] as well as the benefits of multidisciplinary teams.

The pediatric population requires special mention. The tenfold difference in base rate of tracheostomy-related death for children versus adults is disquieting. This finding may reflect the medical complexity of children with congenital disorders, limitations of children in providing self-care, and the narrow caliber lumen of pediatric tracheostomy tubes, making them more susceptible to mucus occlusion. It may also be a reflection of the cumulative risk of tracheostomy-related death in pediatric patients (instead of a higher risk of death per time interval), because the length of time for which the tracheostomy is in place is higher in pediatric patients relative to adults. A study with the National Surgical Quality Improvement Program–Pediatric demonstrated that children younger than 2 years undergoing tracheostomy accounted for the largest contribution to morbidity in otolaryngology, highlighting the need for procedure-specific outcome variables for this group.[14] Although clarifying the

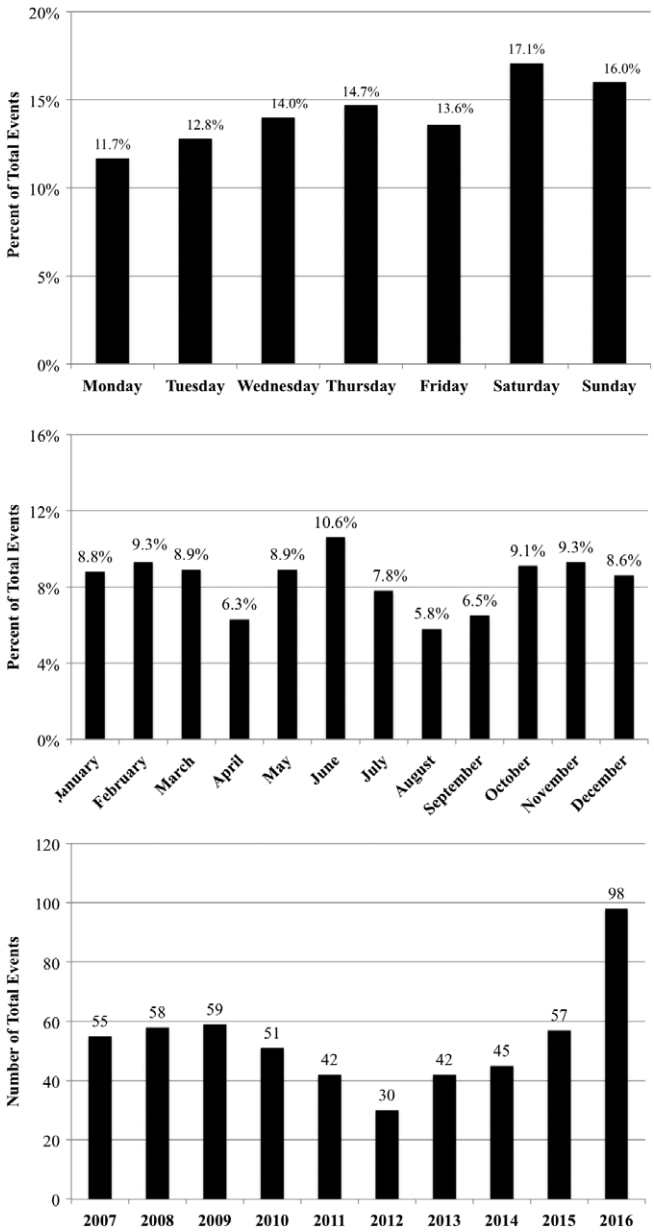

Fig. 2. Day of week, month, and year of deaths related to tracheostomy complications in the adult cohort.

underlying reason is important, it remains clear that nursing needs for these patients are a perennial concern.

Head and neck cancer (HNC) patients, although not analyzed separately from other adult patients due to limited numbers, also deserve special mention. For HNC patients, the underlying base mortality rate following tracheostomy is lower than in critically ill patients.[15] Risk calculators for critically ill patients also predict the risk of mortality following tracheostomy poorly in HNC patients.[15] A separate issue for HNC patients is death due to confusion between a laryngectomy stoma and a tracheostomy with attempted orotracheal intubation,[16] which could not be studied in the present data.

Decreasing inpatient tracheostomy-related complications requires institution-wide protocols for tracheostomy care, implementing simple preventative and rescue measures, and ensuring that staff trained in rescue measures are available at the bedside within minutes of an emergency to intervene before permanent hypoxic injury occurs. Coordinated multidisciplinary teams[17] and protocol-driven, institution-wide efforts involving all providers involved in tracheostomy care drastically reduces tracheostomy-related catastrophic events and mortality. Institutional tracheostomy care improvement efforts in the United Kingdom and Australia reduced tracheostomy-related adverse events by over 90%,[4,5] with similar success replicated in the United States.[18] Within the United States, national organizations have attempted to offer consensus around tracheostomy care best practices,[9] but formal nationally endorsed guidelines are lacking, and most care pathways are institutionally derived. Perioperative care pathways help to ensure that important aspects of care do not fall through the cracks. Patient, caregiver, and nursing staff engagement in these efforts is critical, as the learning curve to achieve comfort and proficiency with tracheostomy care potentially extends beyond 5 years into practice.[19–21] Informed patients and caregivers can ensure tracheostomy tube care, prevent mucus plugging, and seek help early in cases of impending emergency. In addition, development of patient-reported performance measures[22] and other outcome measures can provide patient-centered data that improve assessment of patients with complex upper airways.[23] A patient safety culture also plays a key role, and morbidity and mortality proceedings provide a vehicle for implementing quality improvement efforts around tracheostomy care and promoting Accreditation Council for Graduate Medical Education core competencies.[24]

It is disconcerting that tracheostomy-related deaths were highest in the most recent 2-year period for which data were available. This finding underscores the need for ongoing efforts to improve tracheostomy care standards. Quality improvement collaboratives promote evidence-based enhancements in safety and quality. The Global Tracheostomy Collaborative (GTC) includes 45 member hospitals around the world working to eliminate catastrophic tracheostomy-related events and improving the lives of patients with a tracheostomy. Implementation of GTC drivers decreased length of stay across four National Health Service hospitals.[25] Some authors have suggested that tracheostomy-related catastrophic events might be considered never events, as most events involve absence of preventive/rescue measures or shortfalls in training and staffing.[26] Although some causes of tracheostomy-related mortality, such as tracheoinnominate fistula, are very difficult to prevent, others, such as tracheotomy wounds, have been characterized as never events based on analogy to decubital ulcers.[27] This stance, although controversial, has spawned useful innovations, such as tracheostomy dressing bundles.[28]

Limitations

Our study has several potential limitations. First, as with all large databases, there is potential for error in coding.[29] Second, tracheostomy-related mortality is likely underreported, potentially due to a hesitancy to report

TABLE III.
Multivariable Logistic Regression of Characteristics Associated With Death From a Tracheostomy Complication.

	Children		Adults	
Variable	Odds Ratio	95% CI	Odds Ratio	95% CI
Age				
<2 years	1 (Reference)			
2–5 years	7.76	4.53-13.27	—	—
5–17 years	2.63	1.57-4.42	—	—
18–44 years	—	—	1.78	1.32-2.40
45–64 years	—	—	1.66	1.36-2.04
65–80 years	—	—	1 (Reference)	
80+ years	—	—	0.24	0.17-0.33
Gender				
Female	1 (Reference)		1 (Reference)	
Male	0.96	0.62-1.48	1.18	0.99-1.41
Race				
White	1 (Reference)		1 (Reference)	
African American	2.02	1.26-3.24	1.59	1.29-1.96
American Indian	0.85	0.12-6.14	1.49	0.70-3.14
Asian or Pacific Islander	1.20	0.37-3.88	1.14	0.65-1.98
Ethnicity				
Non-Hispanic	1 (Reference)		1 (Reference)	
Hispanic	1.14	0.64-2.04	1.42	1.06-1.89
Education				
High school or less	—	—	0.86	0.70-1.06
High school or GED	—	—	1 (Reference)	
Some college or associate's degree	—	—	0.80	0.61-1.04
Bachelors, master's, or doctorate	—	—	0.54	0.39-0.75
Marital status	—	—		
Married	—	—	1 (Reference)	
Single, never married	—	—	0.85	0.67-1.08
Divorced	—	—	1.18	0.93-1.49
Widowed	—	—	0.60	0.45-0.81
Location of death				
Decedent's home	1 (Reference)		1 (Reference)	
Hospital, inpatient	1.09	0.59-2.01	4.46	3.40-5.84
Nursing home/long-term care	4.34	0.57-33.31	2.65	1.83-3.82
Other	0.52	0.20-1.39	0.56	0.33-0.96
Day of week				
Weekday	1 (Reference)		1 (Reference)	
Weekend	1.04	0.65-1.64	1.24	1.04-1.49
Month of death				
Spring (March–May)	1 (Reference)		1 (Reference)	
Summer (June–August)	0.87	0.45-1.71	1.06	0.83-1.36
Fall (September–November)	1.31	0.70-2.44	1.08	0.85-1.37
Winter (December–February)	1.75	0.97-3.14	1.05	0.83-1.34
Year				
2007–2008	1 (Reference)		1 (Reference)	
2009–2010	1.40	0.77-2.56	1.00	0.77-1.30
2011–2012	0.61	0.29-1.32	0.65	0.48-0.87
2013–2014	0.63	0.29-1.35	0.77	0.58-1.02
2015–2016	1.41	0.77-2.59	1.31	1.03-1.68

OR > 1 indicates more likely to die from a tracheostomy complication compared to the deceased general population.
CI = confidence interval; GED = general education development test.

iatrogenic complications or possibly due to complacency with what is perceived to be an unmodifiable low-level risk. Third, tracheostomy complications recorded in the data lack specificity, because the tracheostomy complication code does not describe the specific complication that led to death. Thus, we are unable to differentiate between tube dislodgement, mucus plugging, bleeding, or breakdown. It is also unknown what role, if any, confusion with laryngectomy stomas may have played. In addition, the severity of the underlying medical condition and what, if any, relevance it had on tracheostomy-related mortality is not answered by these data. Last, mortality statistics likely only scratch the surface of tracheostomy complications; there are tremendous quality-of-life concerns and overall morbidity related to tracheostomy not reflected in mortality data. Many patients with tracheostomy may experience debilitating anxiety and caregiver burden beyond the scope of this article.[19] The article does not attempt to address the pros and cons of tracheostomy or inform the decision of whether or not to perform one. Rather, it is the first study using comprehensive mortality data to investigate the national incidence of tracheostomy-related mortality in the United States, provides novel data about the scope of the problem, and highlights the need for future work.

CONCLUSION

We report on the largest study to date of tracheostomy-related mortality, including 623 deaths associated with a tracheostomy complication in the United States over a 10-year period. This study provides important perspectives on the scope of the problem and characteristics of individuals affected. Tracheostomy is enormously beneficial for patients with upper airway obstruction or those requiring prolonged mechanical ventilation, and may prevent complications such as ventilator-associated pneumonia. However, once a tracheostomy is placed, it requires meticulous care to be maintained. Deaths directly attributable tracheostomy complications are largely preventable, and such data can provide clues to optimal prevention, education, and rescue strategies. Tracheostomy deaths disproportionately affected younger patients with significant racial and ethnic disparities. The likelihood of tracheostomy-related death also appears higher in healthcare environments, on the weekends, and in less educated patients, reflecting disquieting disparities in healthcare equity and access and highlighting opportunities for quality improvement efforts. This study reaffirms that reducing deaths from tracheostomy complications is a national healthcare imperative.

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